

Animal Identification Via Camera Module Scenario

Project 28 Spring 2025 - Drone System to Monitor Animals

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The **Drone System (DS)** will use a modular camera to scan a set area, sending video to a computer where it's spliced into frames and processed by a Convolutional Neural Network (CNN) detector. The CNN will identify and classify animals with bounding boxes and confidence scores, storing detection data in a database. Over time, users can build animal profiles for the monitored area and make API calls for additional information (size, migration patterns, endangered status, etc.).

Building on Release 1, which established basic animal detection using a laptop camera and YOLOv8 model with Supabase integration, this scenario adds: (1) hardware transition to Raspberry Pi Camera Module (simulating onboard drone camera), (2) multi-animal detection tracking, and (3) autonomous poaching detection with manual confirmation workflow.

Scenario 2: Monitor Wildlife with Multi-Animal Detection and Poaching Alerts

The scenario's key feature is about connecting the camera module to the computer and subsequently the system. We will be using a programmable Raspberry Pi AI Camera as well as a microSD. Using these two hardware parts, we can allow for the camera to send data to the computer which can then stream the images into the CNN detector and run its processes on said video stream classifying any animals.

The user connects the Raspberry Pi Camera Module via USB and runs:

```
python yolo_prediction.py --camera rpi
```

which will then print to the terminal the following messages:

```
Wildlife Monitor v2.0
Initializing Raspberry Pi Camera...
Camera Status: Connected
Connecting to Supabase database...
Database Status: Connected
Loading YOLOv8 model...
Model Status: Loaded
Starting live detection...
```

If the camera fails to connect, the system prints "ERROR: Raspberry Pi Camera Module not found" and exits. If the database is unreachable, it prints "ERROR: Database connection failed. Please verify network connectivity and Supabase credentials" and exits.

Once initialized, the system opens a video feed window and begins printing detection information to the terminal. When multiple animals appear simultaneously, each animal is processed separately by the AI. The system draws separate bounding boxes for each, assigns

unique IDs (T001, T002, etc.), displays species and confidence for each, and saves all detections (if accuracy is above 80%) as separate database entries with the same timestamp. The printout in the terminal will look similar to this:

```
[2025-03-15 14:35:22] Detection ID: T001 | Species: Rhino |  
Confidence: 89% | Saved to database  
[2025-03-15 14:35:22] Detection ID: T002 | Species: Zebra |  
Confidence: 91% | Saved to database.
```

All detections with confidence above 80% are saved as separate database entries with the same timestamp. Detections below 80% confidence are displayed but not saved to the database.

The CNN model has been trained with annotated examples to detect poachers. When the CNN detects potential poaching indicators with confidence above 60%, the system draws a red bounding box around the flagged area in the video feed and prints:

```
[2025-03-15 14:45:30] POACHING ALERT | Confidence: 73% | Queued for  
review  
Pending alerts: 1.
```

The detection is automatically added to a confirmation queue, storing a snapshot, timestamp, and confidence level in a separate poaching_alerts database table. The system continues monitoring without interruption. (Manual review functionality will be implemented in Release 3.)

While monitoring, users can press: 'q' to quit. When quitting with 'q', the system prints:

```
Shutting down Wildlife Monitor...  
=== Session Summary ===  
Session Duration: 00:45:32  
Total Animals Detected: 7  
Species Breakdown:  
- Zebra: 3  
- Rhino: 4  
Poaching Alerts Flagged: 5  
=====
```

Goodbye!.

This scenario passes when: the Raspberry Pi Camera connects and streams video; single and multiple animals are detected and processed separately with correct bounding boxes; detections with 80%+ confidence are saved to the database with species classification; AI autonomously flags potential poaching activity with 60%+ confidence and stores alerts in a separate database table; the system runs continuously for 15+ minutes without errors; session summary displays total detections and species breakdown upon exit.